

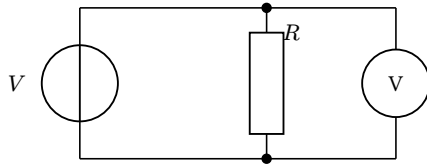
Galeria de figuras

banco de pruebas — no es parte del apunte

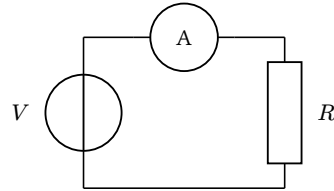
Modulo 1 — Mediciones

`fig-conexion-instrumentos()`

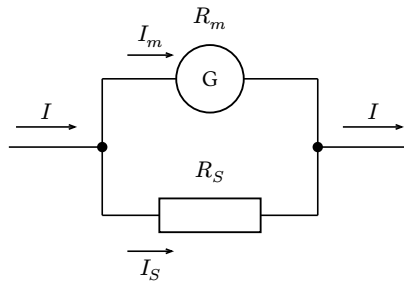
VOLTÍMETRO: EN PARALELO



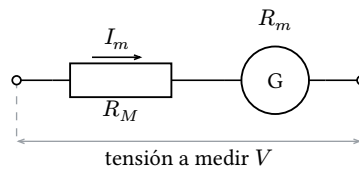
AMPERÍMETRO: EN SERIE



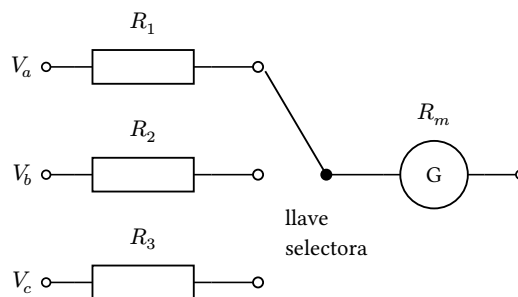
`fig-shunt()`



`fig-multiplicadora()`

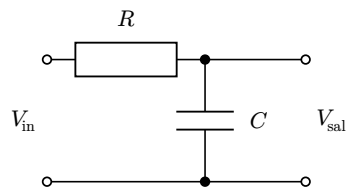


`fig-multirrango()`

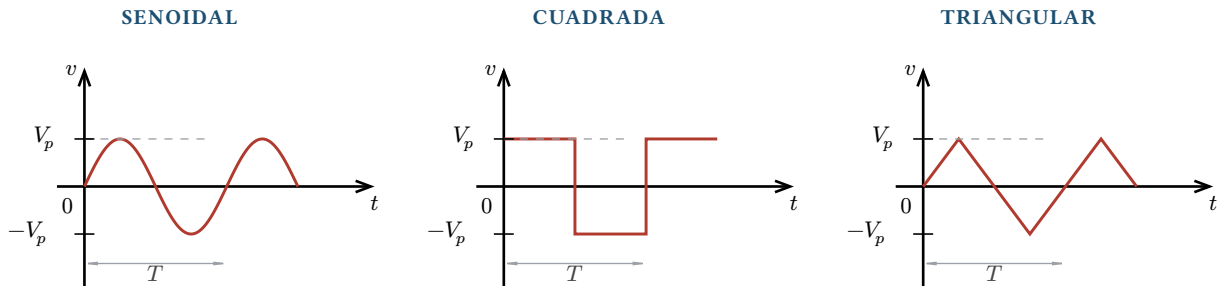


Modulo 2 — Senales

`fig-filtro-rc()`

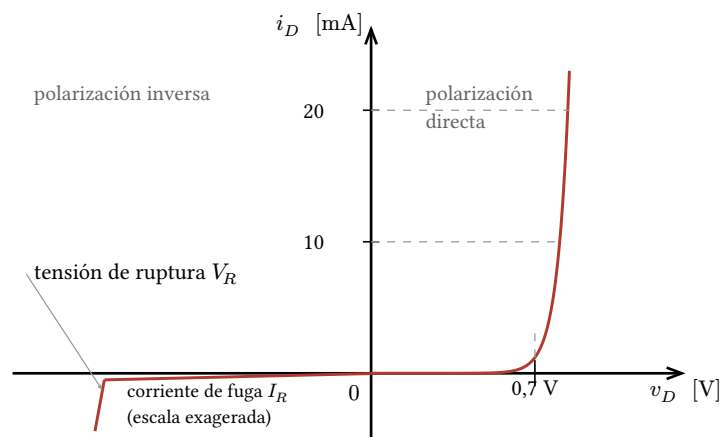


`graf-formas-de-onda()`



Modulo 4 — Diodos

`graf-curva-diodo()`



Modulo 3 — Transformadores

`fig-transformador()`

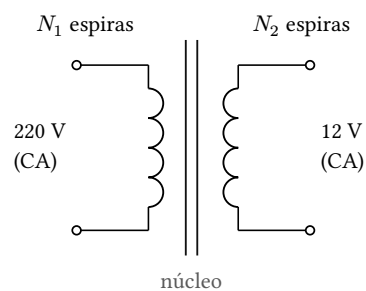
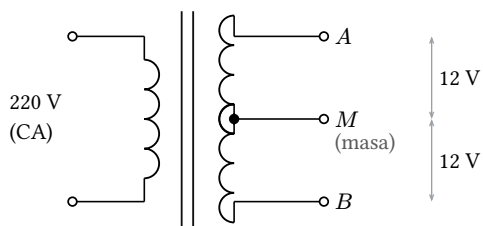


fig-transformador-punto-medio()



Modulo 4 – Diodos (circuitos)

fig-polarizacion-diodo()

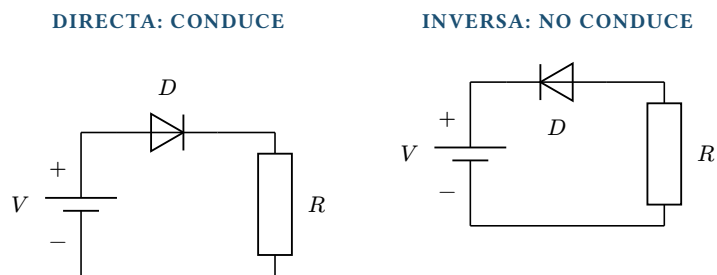


fig-led-limitadora()

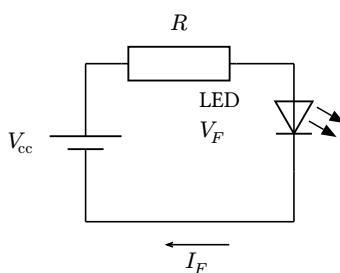


fig-proteccion-polaridad()

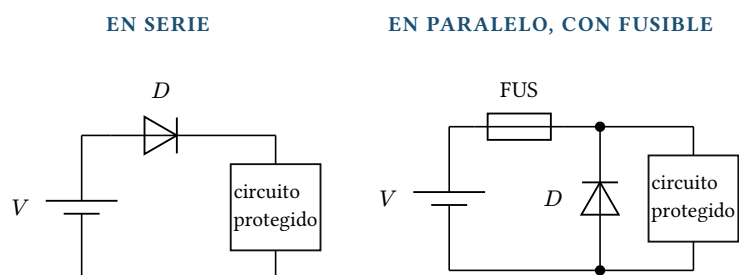


fig-rectificador-media-onda()

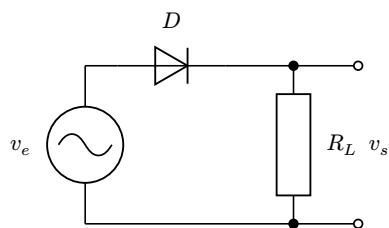


fig-rectificador-punto-medio()

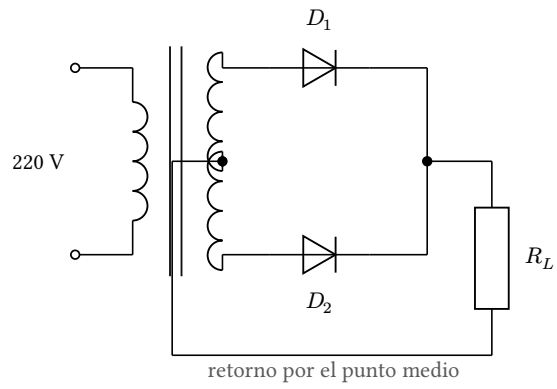
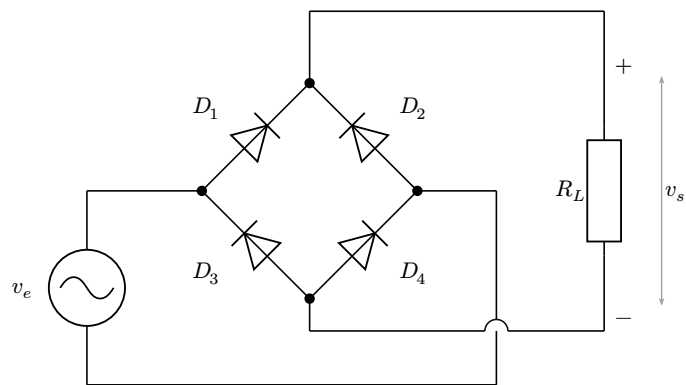
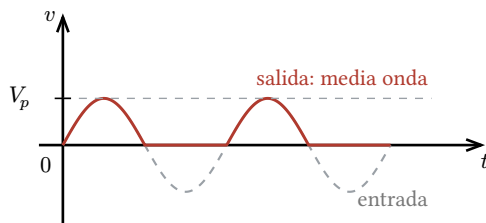


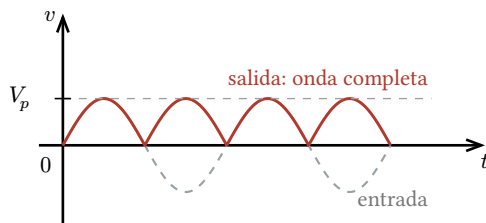
fig-puente-graetz()



graf-media-onda()

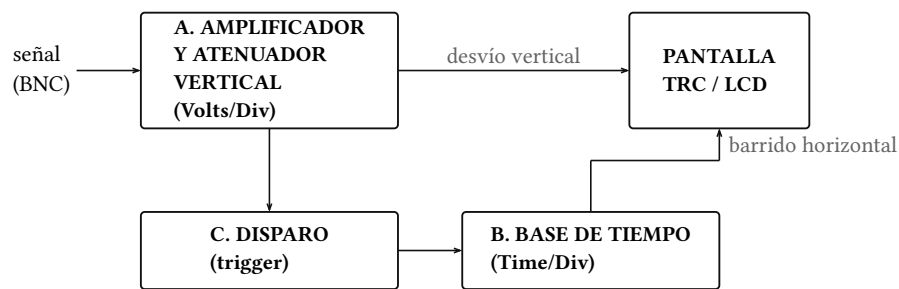


graf-onda-completa()

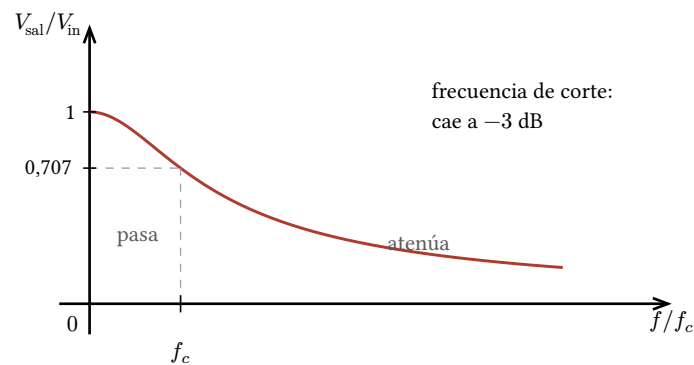


Modulo 2 — Osciloscopio y filtro

fig-bloques-osciloscopio()



graf-respuesta-rc()



Modulo 5 — Fuentes

fig-bloques-fuente()

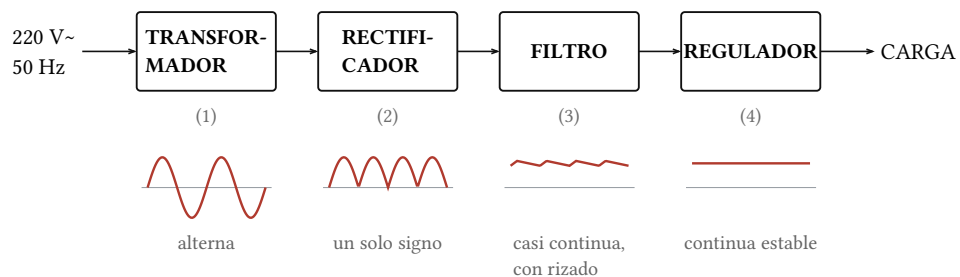
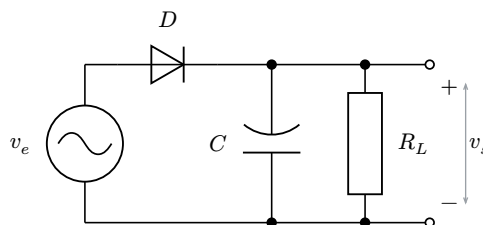


fig-filtro-capacitivo()



graf-rizado()

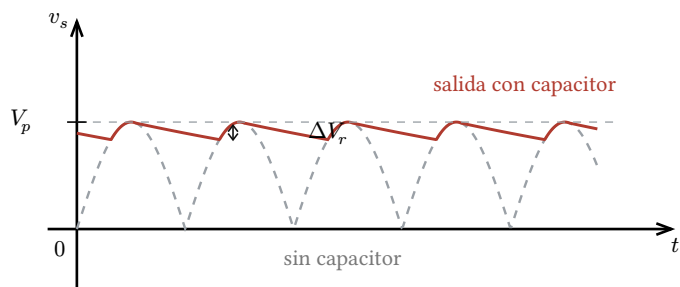


fig-fuente-doble()

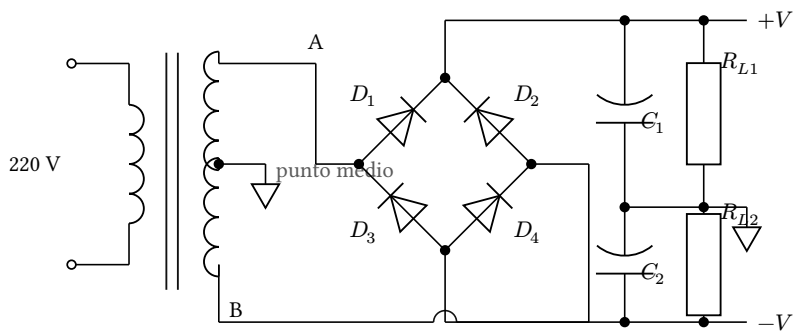
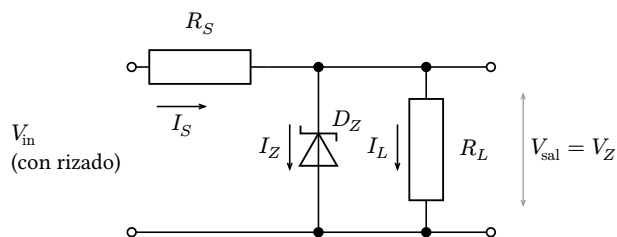
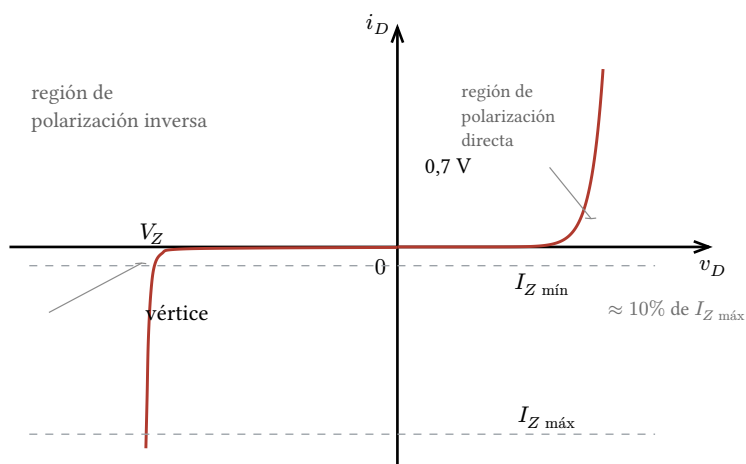


fig-regulador-zener()



graf-curva-zener()



Modulo 6 — Transistores

fig-simbolos-bjt()

NPN — LA FLECHA SALE DEL EMISOR

PNP — LA FLECHA ENTRA AL EMISOR



fig-conmutacion-npn()

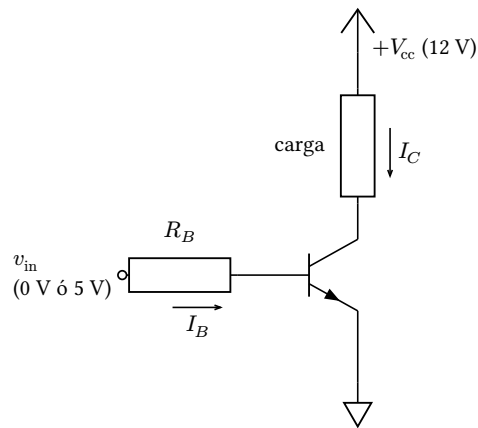
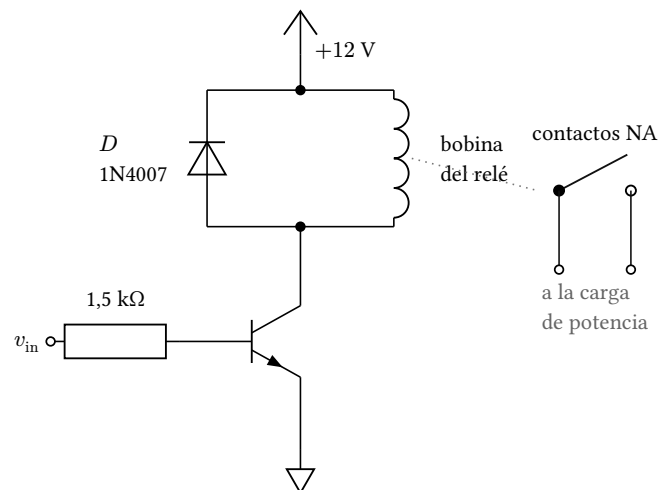
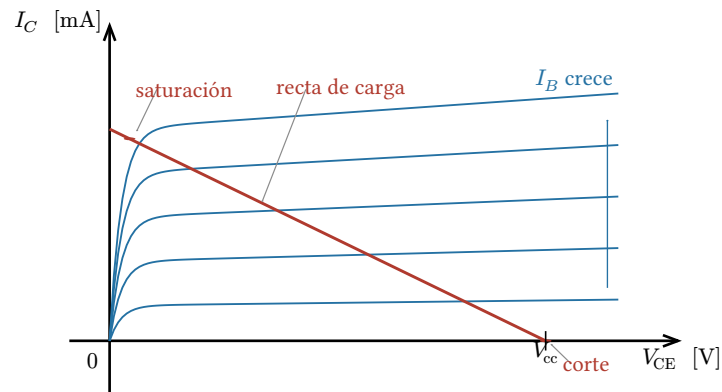


fig-rele-completo()



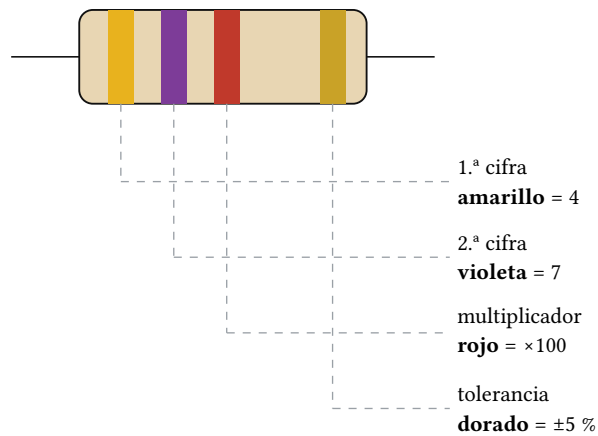
graf-recta-de-carga()



Anexos

fig-codigo-colores()

la tolerancia va separada del resto: por ahí se empieza a leer



$$47 \times 100 = 4700 \, \Omega \pm 5\% = 4k7$$

fig-tabla-simbolos()

resistor	potenciómetro	capacitor	capacitor electrolítico	bobina
diodo	diodo zener	LED	transistor NPN	transistor PNP
fuelle de continua	fuelle de alterna	transformador	lámpara	fusible
llave	voltímetro	amperímetro	galvanómetro	masa

Modulo 7 — Kirchhoff y topologia

fig-nodos-y-mallas()

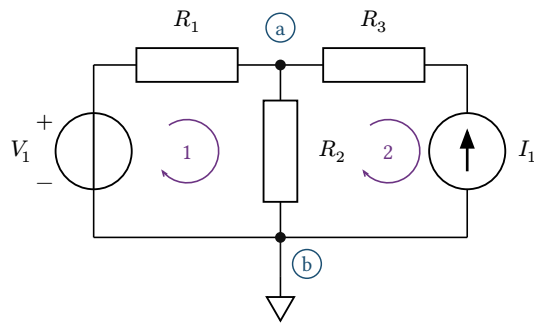
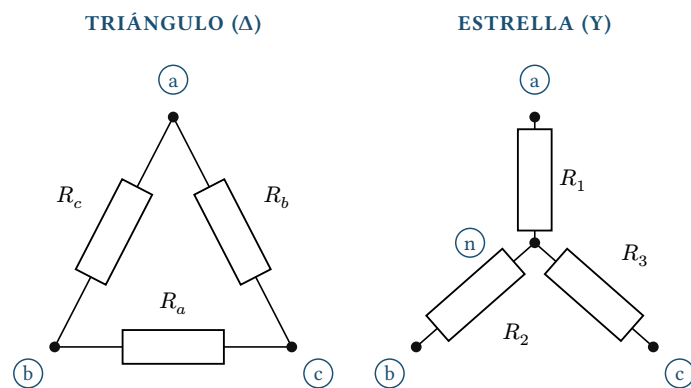


fig-delta-estrella()



Modulo 8 — Nodal y mallas

fig-nodal-primero()

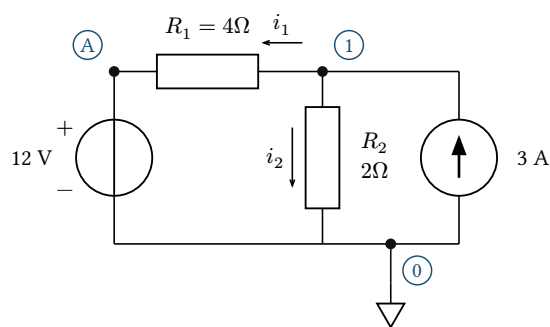


fig-nodal-basico()

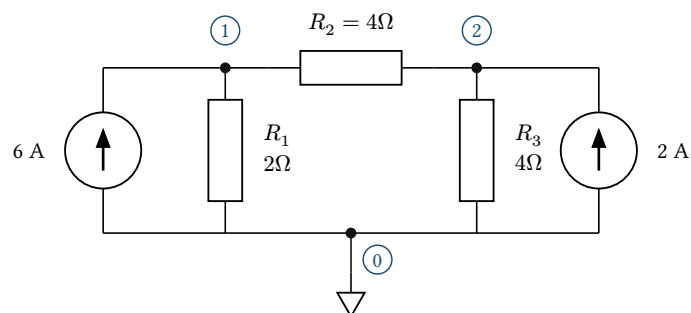


fig-supernodo()

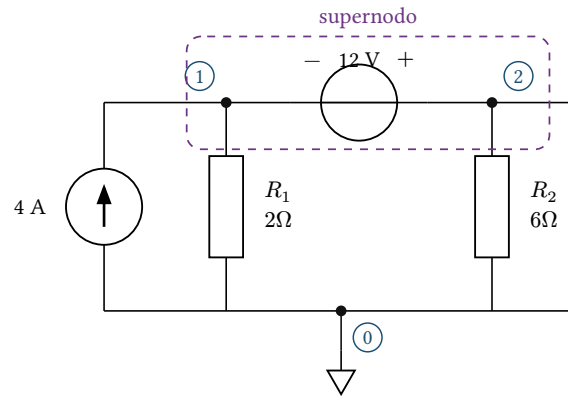


fig-mallas-basico()

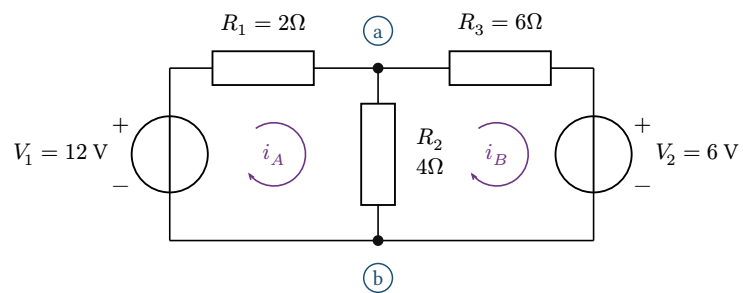


fig-supermalla()

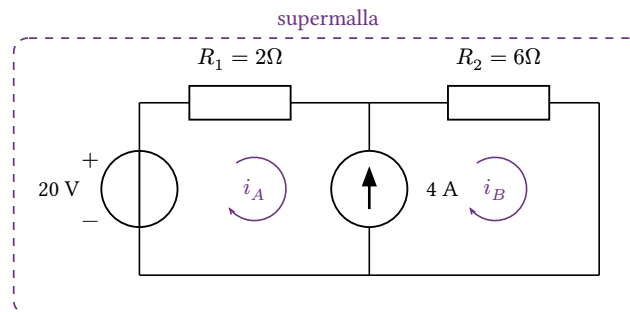
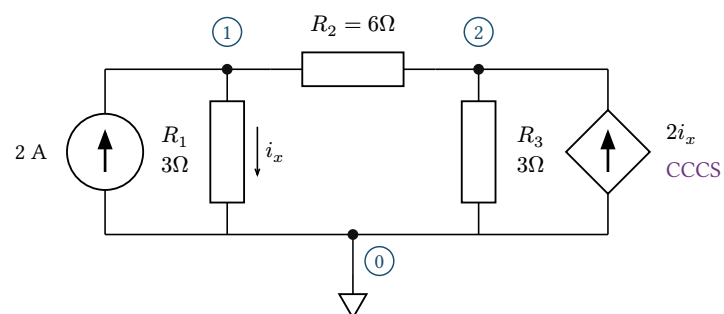
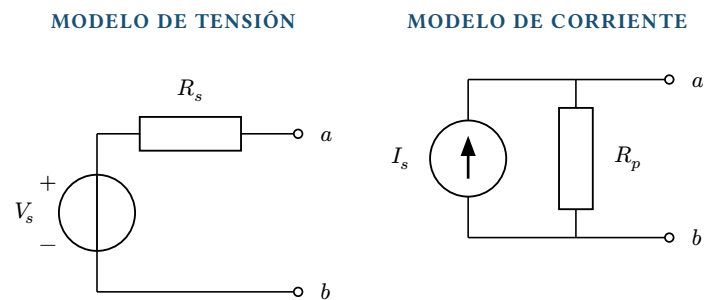


fig-nodal-controlada()



Modulo 9 – Teoremas

fig-fuentes-reales()



Modulo 10 – Transitorios

fig-rc-primer-orden()

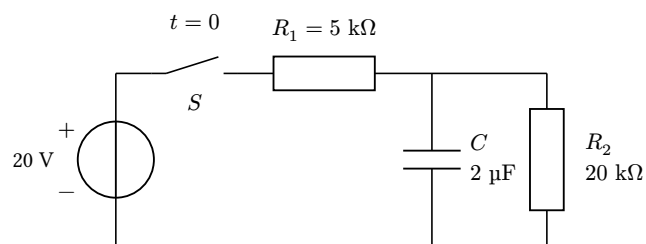


fig-rl-primer-orden()

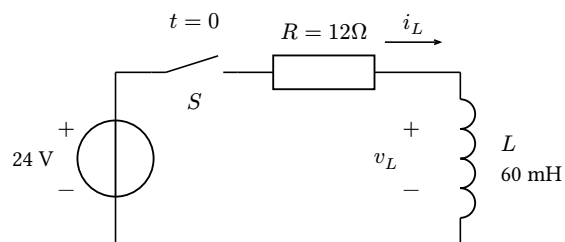


fig-tres-instantes()

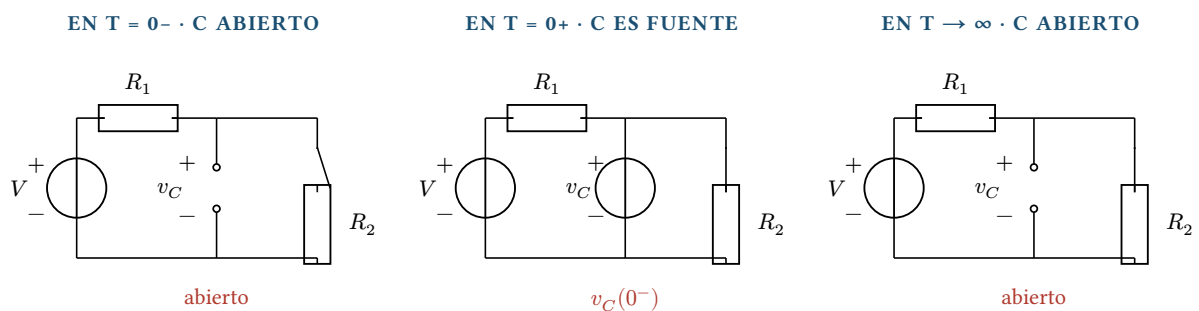


fig-req-prueba()

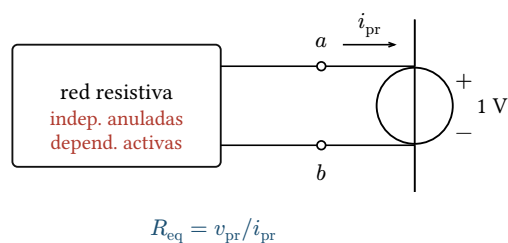


fig-no-idealidades()

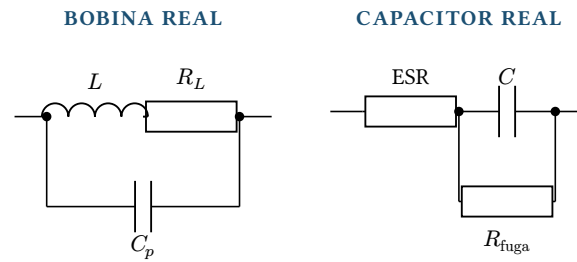


fig-induccion-mutua()

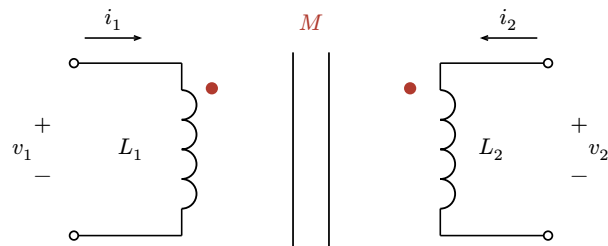


fig-rlc-serie-conmutado()

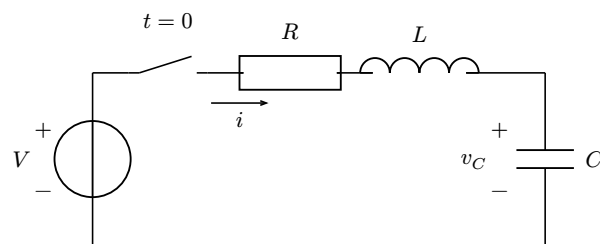
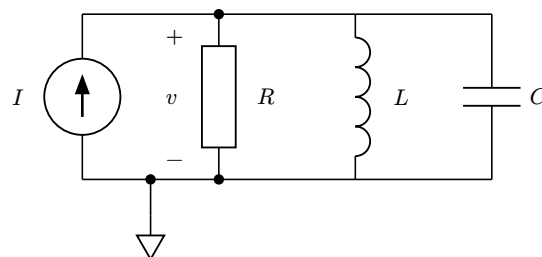
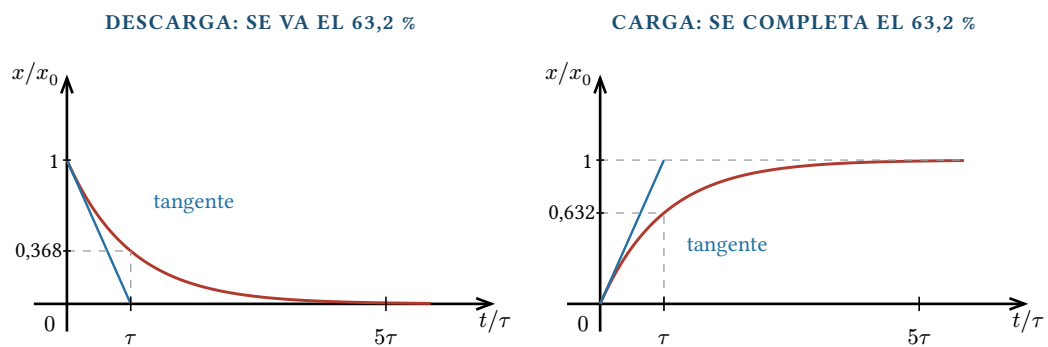
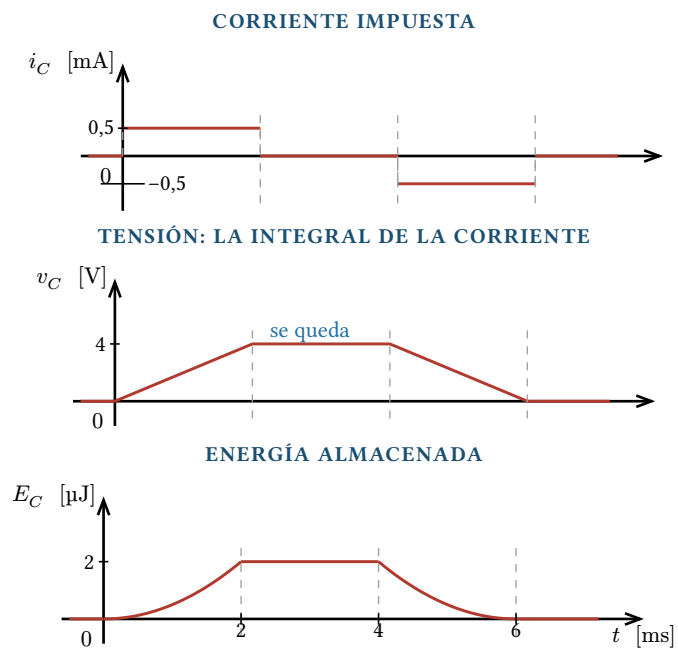
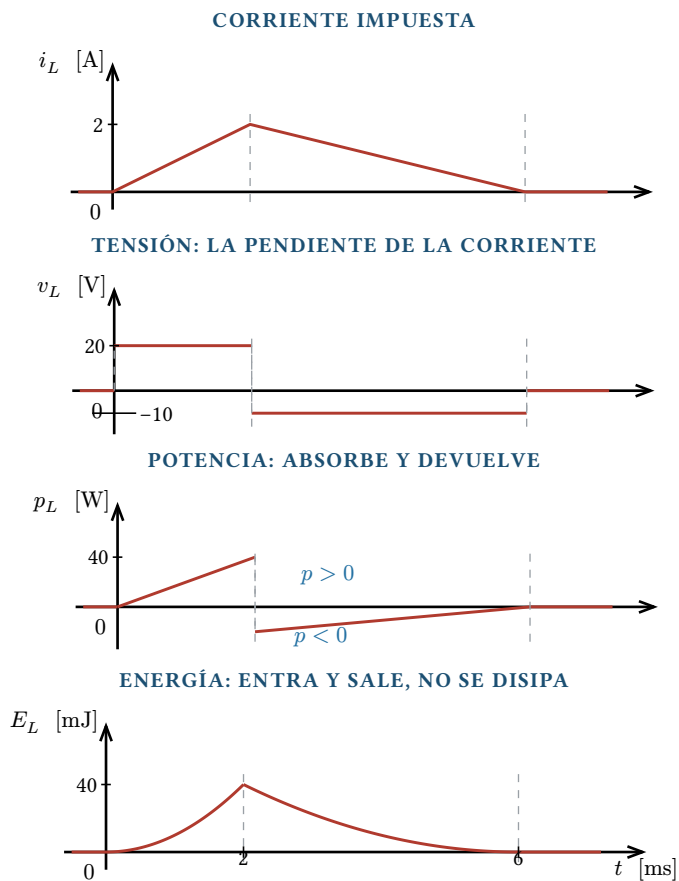


fig-rlc-paralelo()

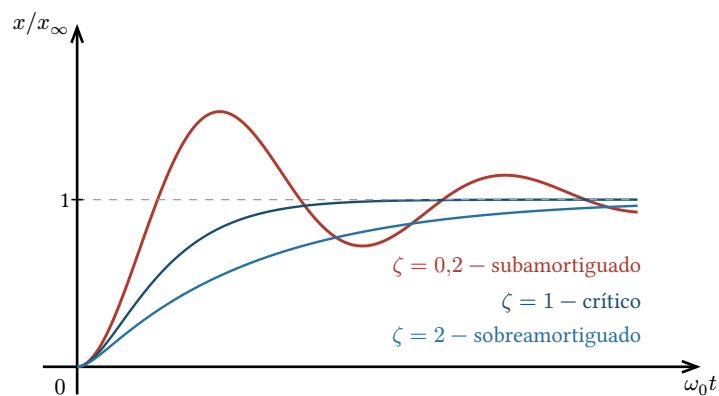


graf-tau-exponencial()

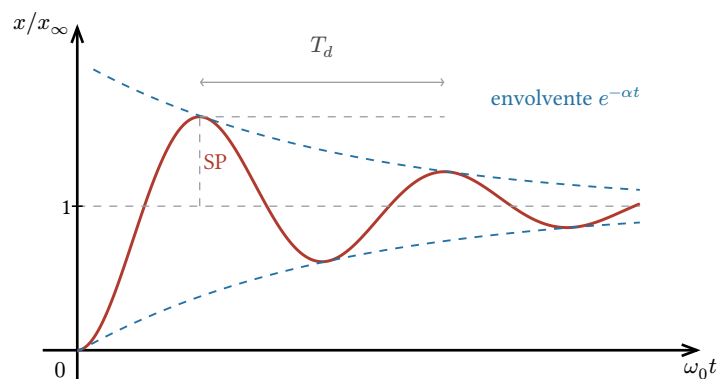




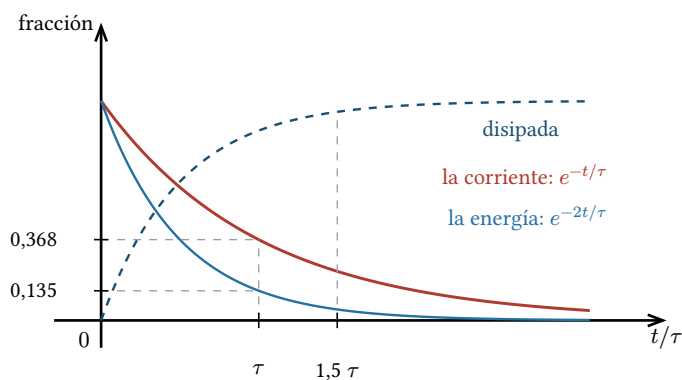
graf-tres-regimenes()



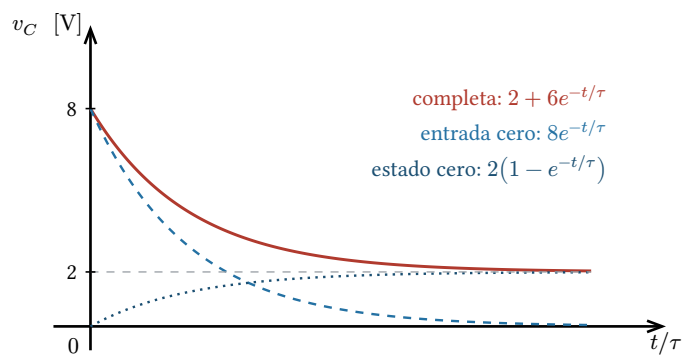
graf-subamortiguado-detalle()



graf-energia-descarga()

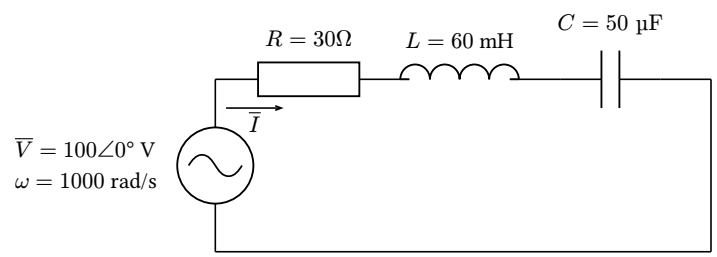


graf-respuesta-completa()



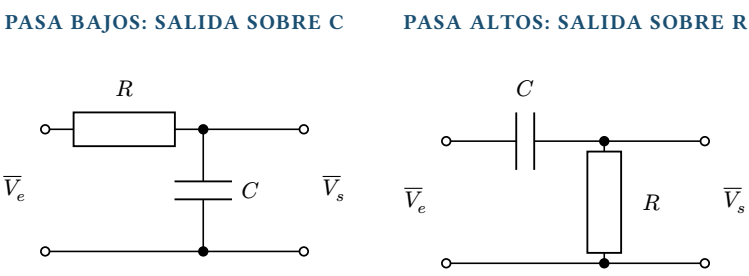
Modulo 11 – Fasores

```
fig-rlc-serie()
```



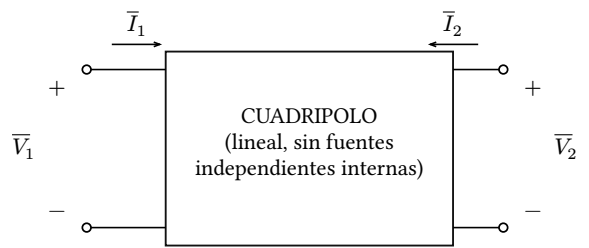
Modulo 12 – Respuesta en frecuencia

```
fig-pasabajos-pasaaltos()
```

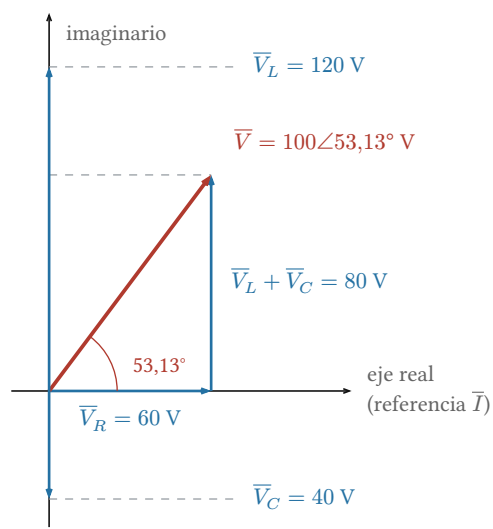


Modulo 13 – Cuadripolos

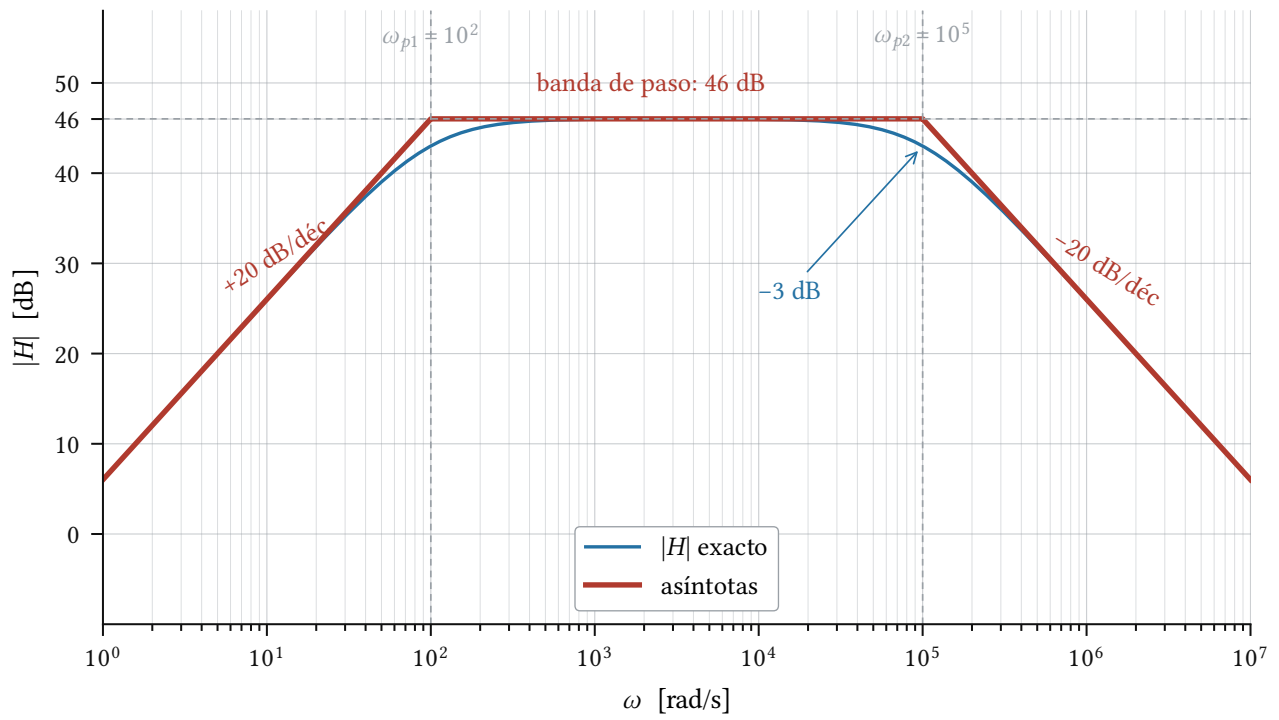
```
fig-cuadripolo()
```



```
graf-diagrama-fasorial()
```



graf-bode-amplificador()



Modulo 14 — El amplificador operacional

fig-ao-terminales()

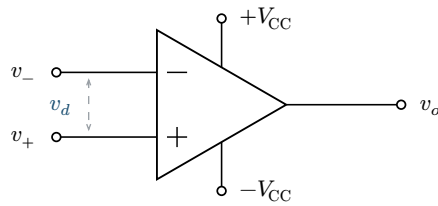


fig-ao-lazo-abierto()

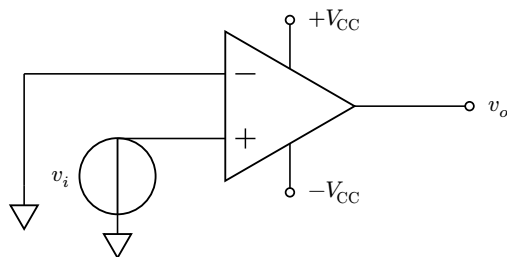


fig-ao-seguidor()

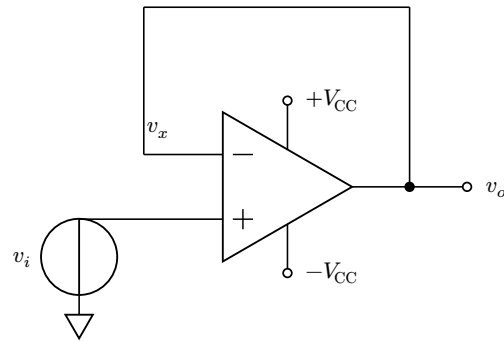


fig-ao-inversor()

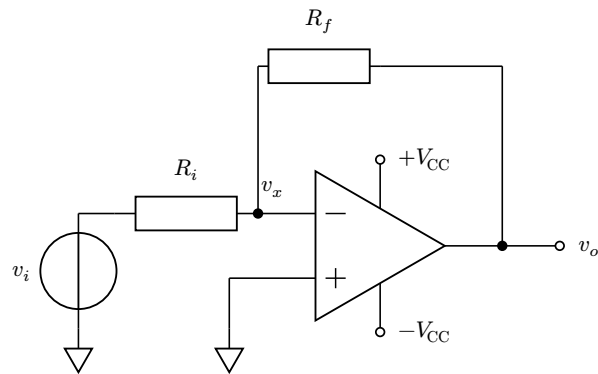


fig-ao-no-inversor()

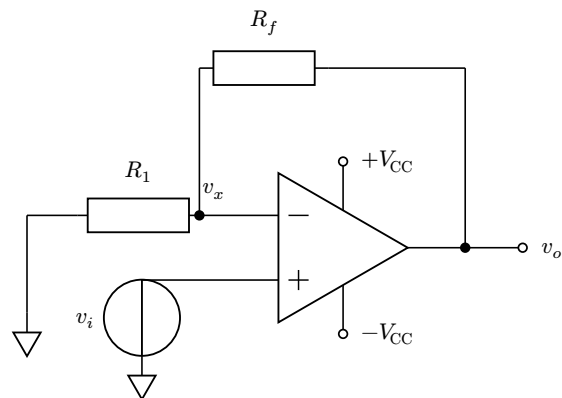


fig-ao-sumador()

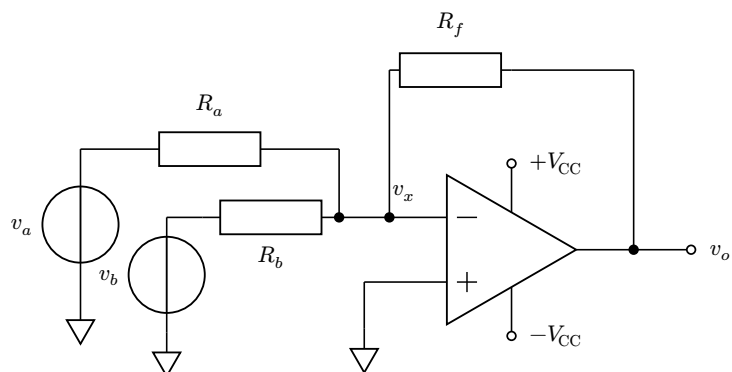


fig-ao-restador()

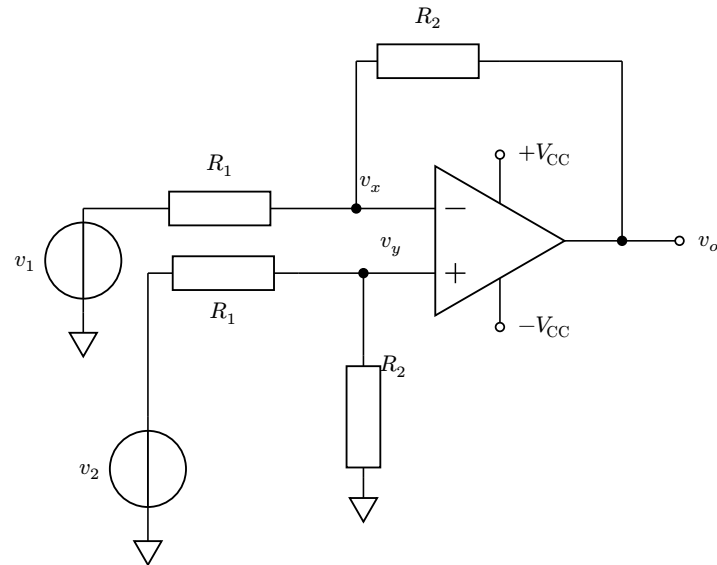
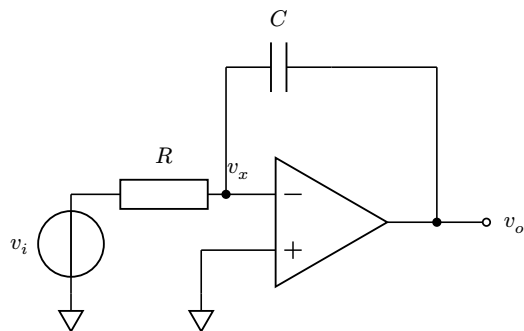


fig-ao-integrador-derivador()

INTEGRADOR: C EN LA REALIMENTACIÓN



DERIVADOR: C EN LA ENTRADA

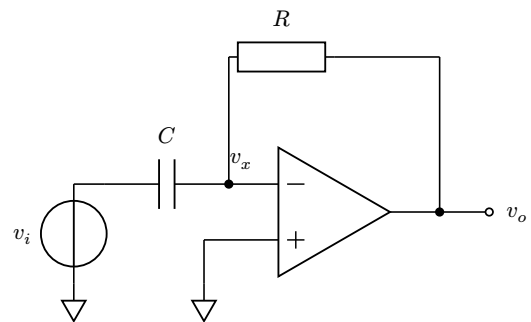
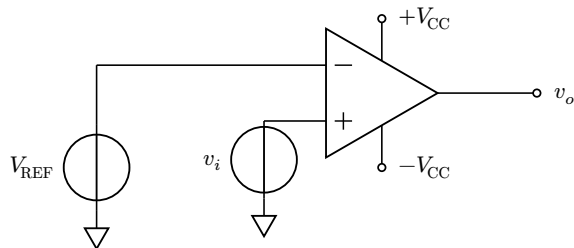


fig-ao-comparador-schmitt()

COMPARADOR: SIN REALIMENTACIÓN



SCHMITT: REALIMENTACIÓN POSITIVA

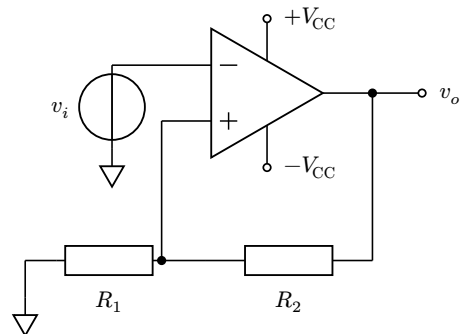
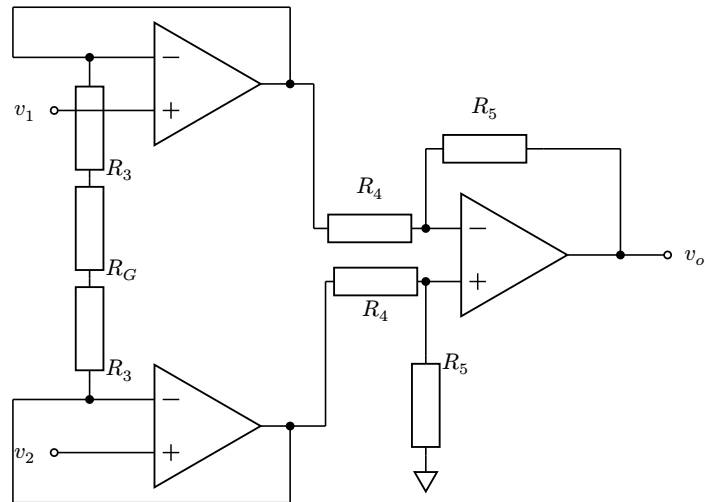


fig-ao-instrumentacion()



Modulo 15 — Simulacion con SPICE

fig-spice-rc()

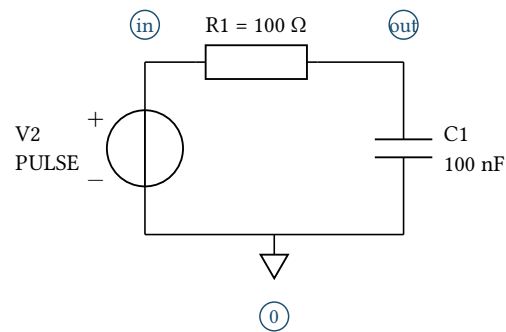
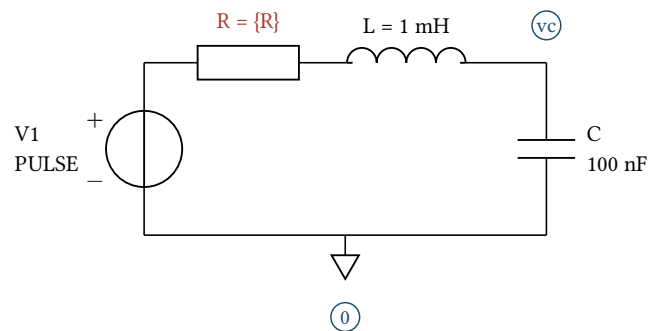


fig-spice-rlc()



.step param R list 10 100 632.46 2k

graf-paso-de-simulacion()

